

Front-End TRAINING

AN INTERNSHIP REPORT

Submitted by

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In fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

in

Information Technology

**SARDAR VALLABHBHAI PATEL INSTITUTE OF TECHNOLOGY
VASAD**



Gujarat Technological University, Ahmedabad

April, 2026



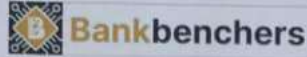
Sardar Vallabhbhai Patel Institute of Technology
Vasad

CERTIFICATE

This is to certify that the internship report submitted along with the internship entitled Frontend Web Development **Training** has been carried out by **Meet Patel** under my guidance in fulfilment for the degree of Bachelor of Engineering in Information Technology, 8th Semester of Gujarat Technological University, Ahmedabad during the academic year 2025-26.

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INTERNSHIP COMPLETION LETTER

Date: 2 April 2026

To Whom It May Concern,

This is to certify that **Meet Patel** has successfully completed his internship as a **Software Engineer Trainee Developer** in the **Frontend Development Department** at **BankBenchers** from **1 January 2026** to **31 March 2026**, under the guidance and supervision of **Mr. Vivek Singh Bhadauria**.

During the internship period, he was engaged in developing responsive web interfaces using React and Next.js, building dynamic user experiences, and integrating frontend components with backend APIs. He demonstrated sincerity, dedication, and a strong willingness to learn, and performed his assigned responsibilities satisfactorily.

We appreciate his efforts and wish him success in his future endeavors.

Sincerely,

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DECLARATION

We hereby declare that the Internship report submitted along with the internship entitled Frontend Web Development **Training** submitted in fulfilment for the degree of Bachelor of Engineering in Information Technology to Gujarat Technological University, Ahmedabad, is a bonafide record of original project work carried out by me at **Bankbenchers (24*7 Moneyworks Consulting Pvt. Ltd.)** under the supervision of **Mr. Vivek Singh** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of the Student

Meet Patel

Sign of Student

ACKNOWLEDGEMENT

I extend my heartfelt gratitude to all those who have been instrumental in making my Frontend Web Development internship a valuable and enriching experience. Their constant support, guidance, and encouragement have played a significant role in enhancing my learning and professional development.

I would like to express my sincere appreciation to **Mr. Vivek Singh** for his expert guidance and continuous support throughout the internship. His insights into design thinking, user experience principles, and practical approaches greatly helped me in improving my skills and understanding of real-world projects.

I am equally thankful to **Mr. Ujjwal Agrawal** for his mentorship and valuable suggestions. His constructive feedback and encouragement motivated me to refine my designs and approach problems more creatively and efficiently.

I would also like to express my sincere gratitude to my internal guide, **Prof. Paresh Patel**, and Head of Department, **Dr. Mala Mehta**, along with my college and respected faculty members, for providing me with this opportunity and for their constant encouragement and support during my internship. Their guidance and interest in my work have been truly motivating. This internship has been a transformative learning experience, and I am sincerely grateful to everyone who contributed to its success. Thank you for being a part of my journey.

Submitted By

Meet Patel

ABSTRACT

This internship focuses on the design and development of a **Farmer Crop Sell Bid System**, a web-based platform aimed at connecting farmers directly with buyers through a digital bidding mechanism. The primary objective of the system is to eliminate intermediaries, improve price transparency, and provide better market access for farmers.

During the internship, modern web technologies such as **Next.js, React.js, Redux Toolkit, and REST APIs** were used to build a scalable and user-friendly frontend. The project involved developing multiple modules including authentication (signup/login), crop listing with image and document upload, bidding functionality, auction and fixed-price models, payment flow, and order management.

Key features implemented include dynamic routing, state management using Redux, form validation, responsive UI design based on Figma, and seamless frontend-backend integration. Advanced functionalities such as error handling with toast notifications, pagination, and initial exploration of real-time communication using WebSockets were also incorporated.

The system supports both buyer and seller interfaces, enabling users to list crops, place bids, accept offers, and complete transactions efficiently. Emphasis was placed on performance, scalability, and user experience throughout development.

This project provided practical exposure to real-world application development, enhancing skills in frontend architecture, API integration, problem-solving, and understanding of fintech-based solutions in the agricultural domain.

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List of Abbreviations

MOQ	Minimum Order Quantity
MSG91	A cloud communication platform for SMS/OTP delivery (India)
NEFT	National Electronic Funds Transfer
OTP	One-Time Password

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CHAPTER 1: OVERVIEW OF THE COMPANY

1.1 History

24X7 Moneyworks Consulting Pvt. Ltd is a pioneering fintech and consulting firm focused on driving transformation across rural finance, banking, and startup ecosystems. Under our fintech brand BankBenchers, we partner with financial institutions, cooperatives, and startups to enable digital transformation, risk management, customer-led innovation, and sustainable growth through people-centric design and technology.

Our core strength lies in building and deploying high-impact platforms such as eKisanCredit (eKCC) — a full-fledged digital loan origination system tailored for cooperative banks, PACS, and RRBs — designed end to end for integration with land records, Aadhaar/KYC systems, Loan management system, core banking, credit bureaus and ePACS. We have developed solutions like Ghar Ghar Kisan Abhiyaan Portal for KCC Saturation campaign monitoring and an Innovative Interest Subvention Platform, which streamlines interest subvention claims in real time under government infrastructure schemes.

1.2 Different product / scope of work

- The **eKCC Portal** is a unified digital platform designed to simplify the entire Kisan Credit Card (KCC) loan process.
- It covers the complete journey — from **application to loan disbursement**.
- The platform is built for **accessibility and efficiency**, ensuring ease of use for farmers.
- It provides a **seamless, multilingual, and mobile-friendly interface**.
- Farmers can **apply for and manage their KCC loans digitally** through the portal.
- The system supports **end-to-end digital KCC operations**.
- It integrates with key APIs such as:
 - Aadhaar
 - eSign
 - PAN
 - SMS
 - Land Verification

- The portal is actively onboarding farmers across **multiple regional rural and cooperative banks in India**.
- It offers **real-time dashboards and MIS tools** for effective monitoring and reporting.

1.3 Organization chart

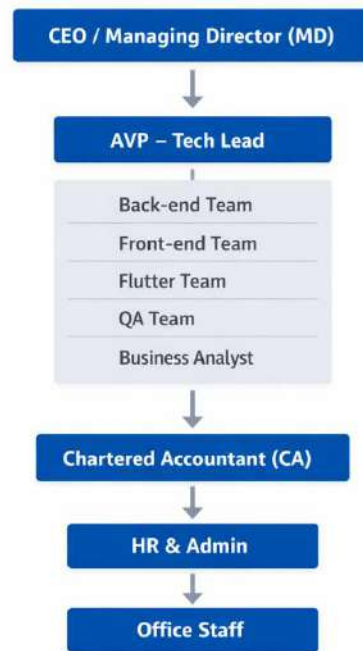


Figure 1.1: Organization chart

1.4 Capacity of plant

BankBenchers is a growing software solutions company with its main headquarters located in Vadodara, India. The organization operates with a focused and efficient team structure, bringing together professionals from various domains such as UI/UX design, software development, quality assurance, and business analysis.

With a strong and capable workforce, the company effectively manages multiple projects simultaneously, ensuring high-quality output and timely delivery. The organizational capacity is well-balanced, allowing seamless collaboration between teams and enabling the company to innovate and enhance its digital solutions continuously.

CHAPTER 2: OVERVIEW OF DIFFERENT DEPARTMENT OF THE ORGANIZATION

2.1 Work carried out in each department

2.1.1 Graphic Design

At BankBenchers, the design team follows a **user-centered design approach**, focusing on creating intuitive and meaningful digital experiences. The primary goal is to understand user needs and deliver solutions that are easy to use and visually appealing. All web solutions are designed to be **responsive across multiple screen sizes**, while mobile applications are crafted to provide a **platform-specific and seamless user experience**. Tools like Figma are extensively used for wireframing, prototyping, and high-fidelity UI design.

2.1.2 Software Development

Software development plays a crucial role in delivering scalable and efficient digital solutions at BankBenchers. The development team works on building systems that are aligned with business requirements and user expectations. Emphasis is placed on creating **robust, secure, and high-performance applications** that support banking and financial workflows.

2.1.3 Website Development

The web development team at BankBenchers builds **interactive, feature-rich, and scalable web applications**. The team integrates APIs, manages databases, and ensures smooth system performance. By using modern technology stacks, developers create solutions tailored to specific business and user needs.

2.1.4 Flutter App Development

BankBenchers utilizes **Flutter** for developing cross-platform mobile applications for both Android and iOS. The focus is on building efficient, high-performance applications that provide a consistent user experience across devices. These applications are designed keeping usability, performance, and scalability in mind.

2.2 Technical Specifications

The technical environment at BankBenchers includes a variety of modern tools and technologies. Visual Studio Code is commonly used for front-end development, while IntelliJ IDEA is used for back-end development tasks.

For web development, technologies such as HTML, CSS, JavaScript, Angular, and Bootstrap are utilized. UI/UX design processes are carried out using Figma, which helps in designing, prototyping, and collaboration.

Additionally, the organization uses tools like Power BI and Microsoft Dynamics to support business operations, data visualization, and decision-making processes.

2.3 Sequence of operation for developing an end product

BankBenchers follows the **Software Development Life Cycle (SDLC)** as a structured approach to develop high-quality software products. This framework defines all stages involved in the development process, ensuring systematic execution and delivery.

2.4 Details of stage of production.

Analysis

- Define project scope, objectives, and requirements.
- Identify resources, timelines, and constraints.
- Conduct feasibility analysis to evaluate project viability.

Design

- Create system architecture and design plans.
- Design user interfaces, workflows, and system components.
- Develop wireframes and prototypes for better visualization and feedback.

Implementation

- Develop the application based on design specifications.
- Perform unit testing for individual components.
- Integrate modules to build the complete system.

Testing

- Conduct functional, performance, and usability testing.
- Identify and resolve bugs and errors.
- Ensure the product meets quality standards and user expectations.

Deployment

- Prepare the application for release.
- Deploy the system to production environments.
- Provide necessary support and training to users.

Maintenance

- Fix issues identified after deployment.
- Update the system to adapt to new technologies and requirements.
- Improve features based on user feedback.
- Ensure continuous performance monitoring and support.

CHAPTER 3: INTRODUCTION TO INTERNSHIP

3.1 Internship Summary

During this internship, I underwent mentored training focused on enhancing my frontend development skills, logical thinking, and problem-solving abilities. I was exposed to a variety of tasks with different levels of complexity, each presenting unique development challenges. As I progressed, I became more confident in handling these challenges through structured coding approaches and continuous feedback from my mentors.

My role in the organization was as a Frontend Developer Intern, where I gained hands-on experience in modern web development practices. I learned key aspects of frontend development such as component-based architecture, state management, API integration, routing, and responsive design. I worked extensively with technologies like React.js, Next.js, and Redux Toolkit to build scalable and efficient applications.

In addition to learning, I actively contributed to real-world projects, where I developed user interfaces, implemented authentication systems, and integrated frontend applications with backend APIs. I also learned how to maintain code structure, improve performance, and enhance user experience based on real-world requirements.

One of my major contributions during the internship was working on the Farmer Crop Sell Bid System, where I helped develop features such as crop listing, bidding functionality, payment flow, and order management. This experience provided me with a deeper understanding of how digital platforms are built to meet both user needs and business objectives.

Overall, this internship significantly strengthened my interest in web development and helped me build a strong foundation in developing efficient, scalable, and user-friendly web applications.

3.2 Purpose

- To develop and strengthen my frontend development skills by learning and applying modern web technologies, tools, and industry practices.
- To gain practical exposure to a professional corporate environment and understand how real-world software projects are developed and managed.

- To build a clear understanding of my career path in web development and align my skills with industry requirements.
- To understand the roles and responsibilities of a frontend developer, including UI development, API integration, state management, and debugging.

To improve my productivity, problem-solving abilities, and time management skills while working on real-time projects and meeting deadlines.

3.3 Objective

The objective of this internship is to gain practical experience and build expertise in the field of web development through hands-on involvement in real-world projects. The internship focuses on applying modern frontend technologies, tools, and development methodologies to create efficient and user-friendly web applications.

It aims to provide opportunities for learning through mentorship, enabling the development of skills such as UI development, API integration, state management, and responsive design. Additionally, the internship offers exposure to real-world applications, helping in understanding how scalable and efficient software solutions are developed to meet both user needs and business goals.

3.4 Scope

The scope of this internship lies in gaining comprehensive exposure to the field of web development, which plays a crucial role in building modern digital platforms. Web development is widely applicable across various industries such as fintech, e-commerce, healthcare, education, and more, making it a highly in-demand and evolving domain.

During this internship, the scope includes working with modern frontend technologies and development practices to understand the complete application development process—from UI design implementation and component development to API integration and state management. The internship also provides insight into how frontend development integrates with backend systems to create seamless and efficient applications.

With the rapid growth of digital platforms, the demand for skilled web developers continues to increase. Organizations focus on delivering scalable, responsive, and user-friendly applications, making frontend development an essential part of

software development. Tools like React.js, Next.js, and Redux Toolkit further enhance efficiency and performance in modern web applications.

In conclusion, this internship offers a strong foundation in web development, helping in understanding industry practices, tools, and challenges. It prepares interns to contribute effectively to real-world projects and opens up diverse career opportunities in the growing field of software development.

3.5 Internship Planning

3.5.1 Internship Development Approach

The internship was structured into multiple phases to ensure gradual learning and practical exposure to web development. The first phase included company orientation and initial training, where I was introduced to the organization's workflow, development processes, tools, and ongoing projects.

The second phase focused on learning frontend development fundamentals, including React.js concepts, routing, state management, API handling, and responsive design. During this phase, I received mentorship and worked on practice tasks to strengthen my development skills.

The final phase involved working on live projects, where I applied my learning to real-world scenarios. I contributed to developing features such as authentication, crop listing, bidding system, and UI components for the Farmer Crop Sell Bid System

3.5.2 Internship Effort and Time, Cost Estimation

The internship effort, time, and progress were managed on a weekly and daily basis. This included tracking working hours, task completion, and deadlines. Daily work logs were maintained to record development tasks such as coding, debugging, API integration, and UI implementation.

The internship was primarily focused on learning and practical exposure, and the efforts were evaluated based on consistency, quality of work, and timely delivery of assigned tasks.

3.5.3 Roles and Responsibilities

During my internship, my role was that of a Frontend Developer Intern. My key responsibilities included understanding project requirements from mentors and translating them into functional and efficient web applications.

I worked on developing UI components, integrating APIs, managing application state, and implementing responsive designs while ensuring performance and usability. I collaborated closely with team members, including developers and mentors, to align development outputs with project requirements.

Additionally, I was responsible for maintaining daily progress logs, incorporating feedback, and continuously improving my coding approach based on best practices and development standards.

3.5.4 Group Dependencies

During my internship at 24×7 Moneyworks Consulting Pvt. Ltd., collaboration and coordination across teams played a crucial role in delivering effective solutions. I worked closely with backend developers and mentors to ensure that frontend development aligned with API structures and system requirements.

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Regular interactions with mentors and team members through discussions and feedback sessions helped refine my work and improve application performance. This collaborative environment enhanced my ability to adapt, communicate effectively, and work efficiently within a professional development team.

CHAPTER 4: PLATFORM ANALYSIS

4.1 Project

KisanSetu (meaning 'Farmer Bridge' in Hindi) is a web-based agricultural marketplace platform that bridges the gap between farmers and buyers through a clean, accessible, and intuitive digital interface. From a development perspective, the platform is designed as a modern web application — ensuring that it remains fast, responsive, and usable across devices, even in low-bandwidth rural environments.

The platform presents two distinct trading experiences within a single unified interface:

- Whole Lot Auction Interface — A real-time, dynamic bidding screen where live bid updates, countdown timers, and bid history are displayed without page reload, creating an engaging and transparent auction experience for farmers.
- Partial Fixed-Price Order Interface — A straightforward product-style browsing and ordering flow familiar to users of e-commerce platforms, with clear stock availability indicators and a 10-minute checkout timer.

A key feature of the platform is the **role-based functionality**, where users can act as both buyers and sellers within a single account. This simplifies the user experience and eliminates the need for managing multiple accounts.

The platform is designed to serve a wide range of users, including rural farmers using basic smartphones and buyers from different markets. Therefore, usability, responsiveness, and simplicity are considered essential aspects of the system design.

4.2 Objective

From a system development perspective, the objectives of the Farmer Crop Sell Bid System translate into the following functional goals:

1. Eliminate Middlemen — Transparent Information Design: The system enables direct communication and transactions between farmers and buyers, reducing dependency on intermediaries and ensuring better profit margins for farmers.

2. **Fair Price Discovery — Real-Time Visual Feedback:** The auction-based model allows buyers to place bids openly, ensuring fair competition and helping farmers receive optimal prices for their crops.
3. **Secure Payments — Digital Platform Accessibility** The platform ensures secure transactions through structured workflows and clear transaction steps, providing trust and reliability to both buyers and sellers.
4. **Expand Market Access — Mobile-First Responsive Design:** The system allows farmers to access a wider market digitally, enabling them to reach multiple buyers beyond their local region.
5. **Synchronize Demand and Supply — Intuitive RFQ Flow:** The crop listing and bidding system enables better alignment between buyer demand and farmer supply, improving overall market efficiency.
6. **Build Trust — Consistent Visual Language:** Consistent use of status indicators, badge colors, notification styles, and action buttons across all modules creates a predictable interface that builds user confidence over time.

4.3 Scope

- **Registration & Login Screens —** Clean onboarding flow with OTP verification, inline validation, and clear error messages that guide users without confusion.
- **Unified Dashboard —** Separate Sell Mode and Buy Mode stat cards, notification bell, recent activity feed, and Quick Action buttons — all accessible from one screen with a toggle switch.
- **Listing Creation Flows —** Step-by-step form design for both Whole Lot Auction and Partial Order listing types, including photo upload with drag-and-drop support and live preview.
- **Auction Bidding Interface —** Real-time bid ticker, current highest bid display, countdown timer, bid history list, anti-snipe extension alerts, and bid placement form with validation.
- **Partial Order Checkout —** Product detail page, quantity selector with MOQ enforcement, 10-minute reservation countdown timer, and payment initiation screen.

- Escrow Status Tracker — Step-by-step escrow progress UI visible to both buyer and seller with clear CTAs at each stage.
- RFQ Module — Requirement posting form for buyers and offer comparison table with farmer details, price, quantity, and dispatch date for easy decision-making.
- In-Platform Messaging — Contextual chat interface with listing/order reference, image support, system message styling, and an inline Proceed to Escrow CTA button.
- Dispute Filing Screen — Structured form with reason selection, description field, and evidence upload with clear guidance on what constitutes valid evidence.
- Admin Panel (/admin) — Separate dashboard with overview stats, alert panel, listing moderation queue, dispute resolution interface, user management table, and auction monitoring screen.

CHAPTER 5: SYSTEM REQUIREMENT

5.1 Development Tools

As a frontend developer, the following tools and technologies are directly relevant to the development, implementation, and integration of the Farmer Crop Sell Bid System platform:

5.1.1 Design & Prototyping Tools

Figma

The primary design tool used for creating wireframes, high-fidelity UI screens, interactive prototypes, and the shared component library (Design System) for KisanSetu. Figma's real-time collaboration feature allows designers and developers to work simultaneously on the same file, reducing handoff friction. The complete prototype is published and accessible at kisansetu.figma.site.

Component Library / Design System

A shared design system is maintained in Figma covering:

- Color palette (primary, secondary, success, warning, error states)
- Typography scale (headings, body, captions) using Inter font family
- Spacing and grid system (8px base grid)
- Reusable components: buttons, form inputs, badges, status chips, cards, modals, tables
- Icon set for crop types, auction states, escrow stages, and notification types

Responsiveness Testing

- Mobile: 360px – 480px (primary target for farmer users)
- Tablet: 768px – 1024px
- Desktop: 1280px and above

5.1.2 Frontend Implementation Stack

React.js

The frontend is developed using React.js, which provides a component-based architecture for building scalable and maintainable user interfaces. It enables efficient rendering and reusable UI components across the application.

Next.js

Next.js is used for routing, server-side rendering, and overall application structure. It improves performance, SEO, and provides advanced features like dynamic routing and API handling.

Redux Toolkit + RTK Query

Redux Toolkit is used for global state management, including handling user authentication, crop listing data, bidding information, and application state synchronization across components.

REST-API-Integration

Frontend is connected with backend services using REST APIs to perform operations such as authentication, listing management, bidding, and order processing.

Socket.IO Client

Powers all real-time UI updates — live bid amounts on the auction screen, new message indicators in chat, and notification bell badge counts — without requiring page refresh.

5.1.3 Key UI Performance Requirements

UI Requirement	Target Metric
Dashboard load time	Under 2 seconds on 4G
Auction bid update (UI refresh)	Under 1 second from server event
Partial order stock lock API response	Under 200ms (p99)
Message delivery to chat UI	Under 1 second
Minimum supported screen width	360px
Photo upload per listing	Min 2, Max 10 photos (each under 5MB)

Table 5.1: Key UI Performance Requirements and Target Metrics

5.1.4 Infrastructure & Cloud Tools

Component	Tool / Service
Version Control	Git & GitHub
Development Environment	VS Code
Package Manager	npm
API Testing	Postman
Deployment	Vercel/Local Server
State Management	Redux Toolkit
Routing	Next.js Router

Table 5.2: Infrastructure & Cloud Tools Used in KisanSetu

5.1.5 Accessibility & Usability Standards

- All interactive elements are designed to be easily clickable and user-friendly across devices
- Proper contrast and readable font sizes are maintained for better visibility
- Form validation errors are displayed clearly to guide users
- Loading states and feedback messages are provided to improve user experience
- System messages in chat use a distinct visual style separate from user messages
- Simple navigation structure is used to make the platform easy to understand

5.1.6 Hardware Requirements

Development Environment:

- Processor: Intel Core i5 or equivalent (minimum)
- RAM: 8 GB (16 GB recommended)
- Storage: 50 GB free disk space
- OS: Windows 10 / macOS / Ubuntu 20.04+
- Node.js 18+ and Docker Desktop installed

End User (Farmer/Buyer):

- Any Android/iOS smartphone with a modern browser (Chrome, Firefox, Safari)
- Internet connection: 2G/3G minimum; 4G recommended
- No app installation required (Progressive Web Application)

CHAPTER 6: IMPLEMENTATION OF PROJECT

6.1 Client Information Module

In the context of KisanSetu, the Client Information Module refers to the user registration, authentication, and profile management system that captures and manages all information about the platform's two primary client types — Farmers (Sellers) and Buyers.

During registration, the system collects the following information:

- Full Name, Mobile Number (, and Email Address
- State and District (dropdown selection — district populates based on state)
- Password (minimum 8 characters with uppercase, number, and special character)
- Optional: Aadhaar Number (AES-256 encrypted), Bank Account Number and IFSC code

Upon successful registration, the user account is created and authentication is handled through login functionality. The system ensures proper validation and error handling for duplicate entries such as existing email or mobile number.

Development Decisions for this Module:

- Dynamic dropdowns are implemented for state and district selection to improve usability
- Form validation is handled efficiently to prevent incorrect data submission
- Clear and user-friendly error messages are displayed for better guidance
- Login authentication ensures secure access to the platform
- After login, users are redirected to a dashboard where they can access buyer or seller functionalities

6.2 Project Information Module

The Project Information Module represents the core functionality of the Farmer Crop Sell Bid System — enabling farmers to list crops and buyers to browse, bid, or purchase them.

Listing Creation UI (Seller Side):

A farmer in Sell Mode can create two types of listings through a multi-step form:

- Step 1: Crop details (name, variety, grade, harvest date)
- Step 2: Quantity and pricing (with unit type toggle: kg / quintal)
- Step 3: Location (state, district, pincode)
- Step 4: Photo upload (drag-and-drop with thumbnail preview, minimum 2 required)
- Step 5: Review and Submit

A progress bar at the top of the form shows the user which step they are on. Each step validates before proceeding, preventing submission errors.

Auction Bidding Interface:

- Current Highest Bid — Large, bold typography with real-time animated update on new bid
- Countdown Timer — Color-coded (green to yellow to red); displays 'Extended!' banner when anti-snipe triggers
- Bid History Feed — Scrollable list showing masked bidder ID, bid amount, and timestamp
- Bid Placement Form — Input field pre-filled with minimum valid bid amount; Place Bid CTA with loading state
- Listing Photos — Image carousel with zoom support for quality assessment

Partial Order UI:

- Available stock shown as a progress bar (e.g., 320 kg of 500 kg remaining)
- Quantity input with MOQ enforcement (inline error if below MOQ)
- 10-minute countdown timer appears after placing order — visually urgent to drive timely payment

6.3 SURVEY MODULE

The Survey Module in KisanSetu is represented by the Buyer Requirement Posting (RFQ — Request for Quotation) system, which functions as a demand-surveying mechanism on the platform.

Buyer — Post a Requirement Screen:

- Crop Type (searchable dropdown), Required Quantity with unit toggle (kg / quintal / tonne)
- Quality Grade selector (A / B / C / Any) and Target Price (optional)
- Delivery Location (state + district), Delivery Deadline (date picker)
- Offer Deadline (default 72 hours, adjustable) and Additional Notes

Farmer — Browse & Respond to RFQs:

Farmers in Sell Mode see an RFQ listing page with filter options. Each RFQ card shows the crop required, quantity, grade, delivery location, time remaining (color-coded countdown), and a Submit Offer CTA button.

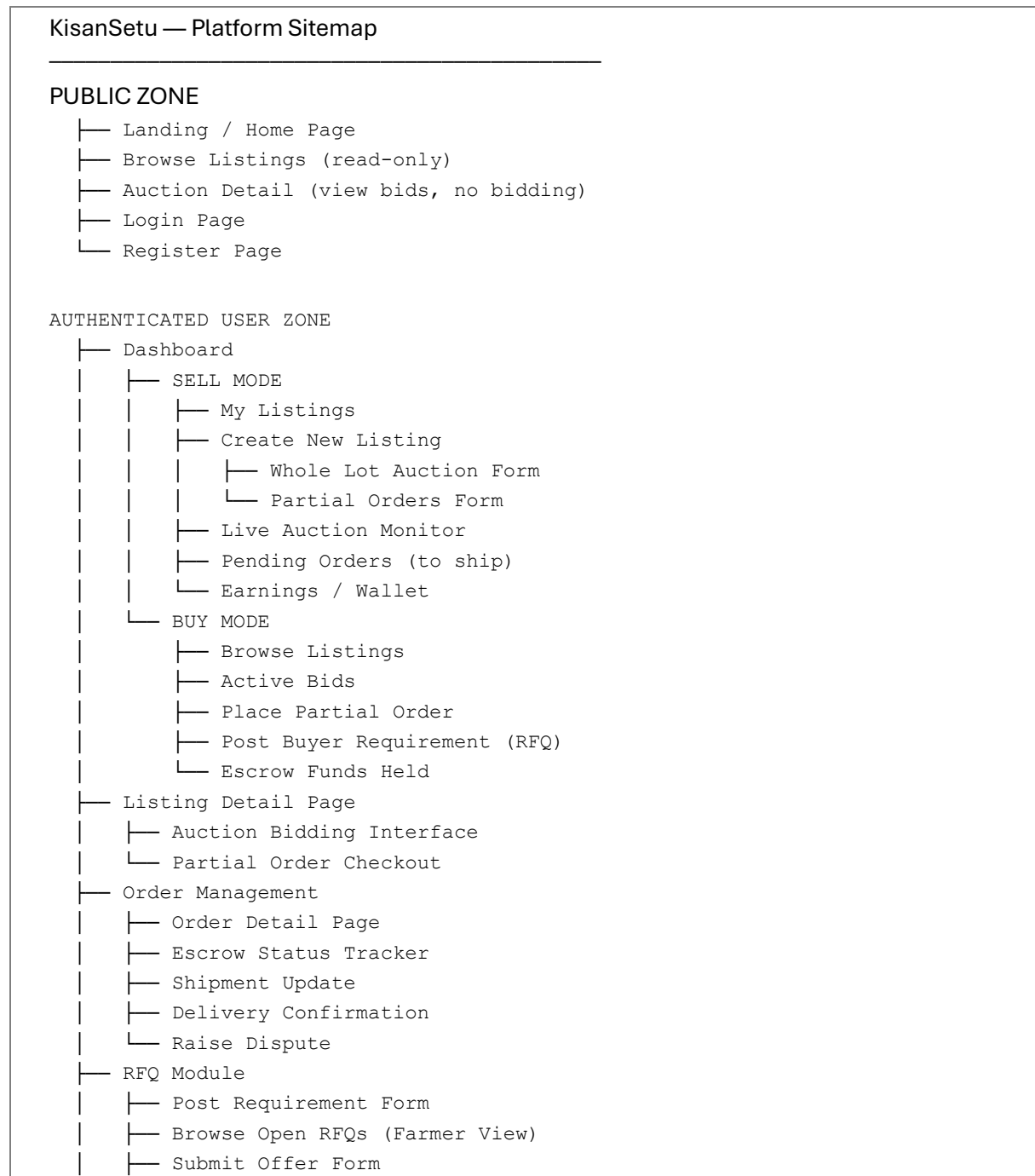
Buyer — Offer Comparison Table:

Once offers arrive, the buyer sees a comparison table with one row per farmer offer showing farmer name, verified badge, state, offered quantity, price per unit, total cost, expected dispatch date, and a Select This Offer CTA. The selected offer is highlighted and the escrow flow initiates automatically.

CHAPTER 7: DESIGN ARCHITECTURE

7.1 Information Architecture (Sitemap)

The Information Architecture of KisanSetu defines how all screens and modules are organized and connected across three access levels — Public, Authenticated User, and Admin. (Figure 7.1)



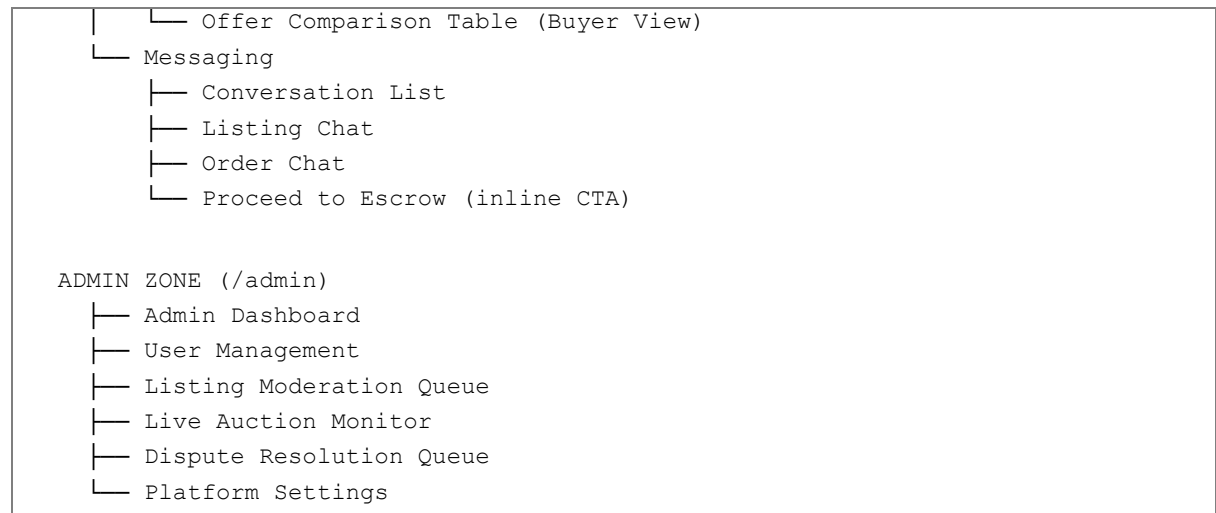
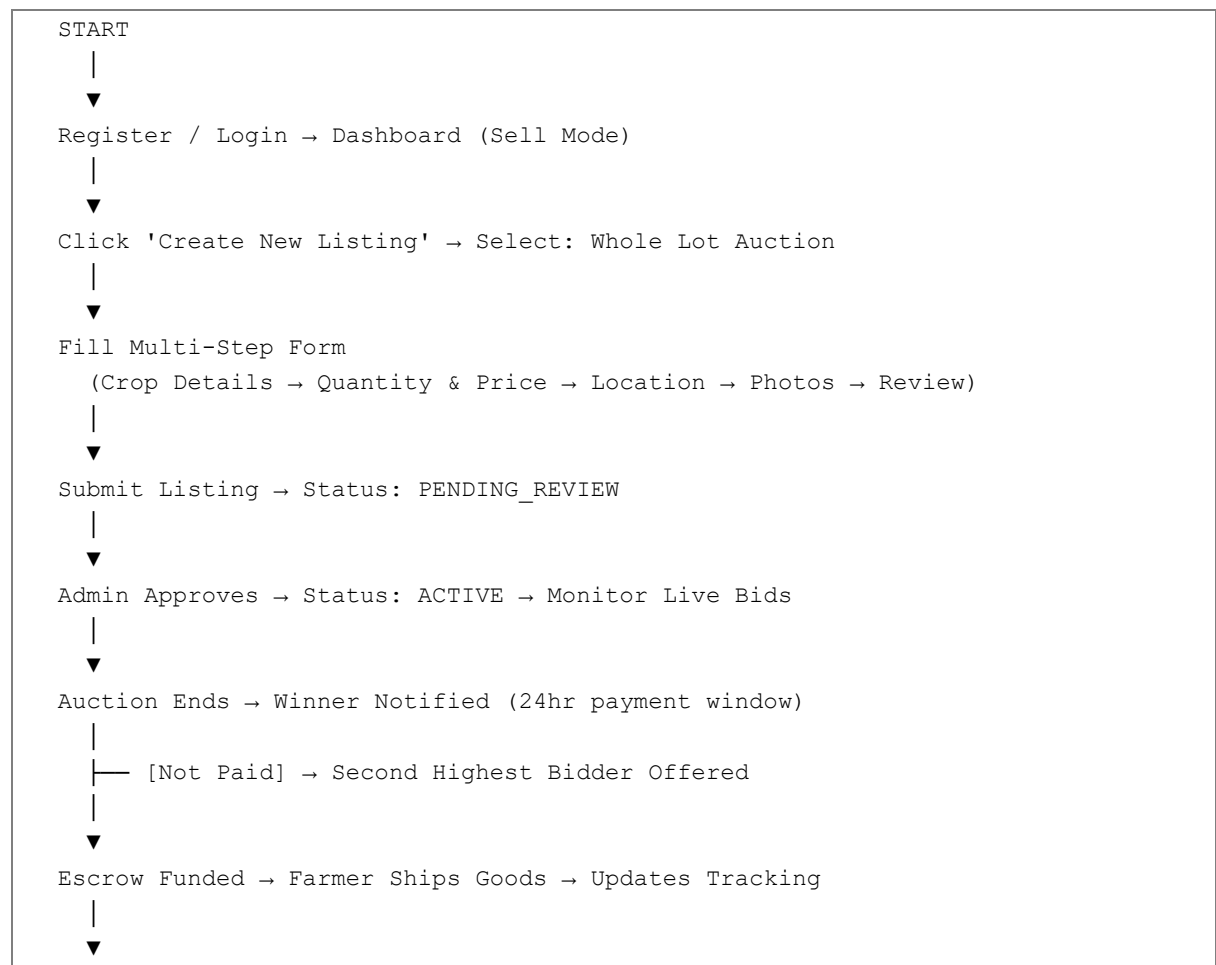


Figure 7.1: KisanSetu Platform Information Architecture (Sitemap)

7.2 User Flow Diagrams

7.2.1 Farmer (Seller) — Whole Lot Auction Flow



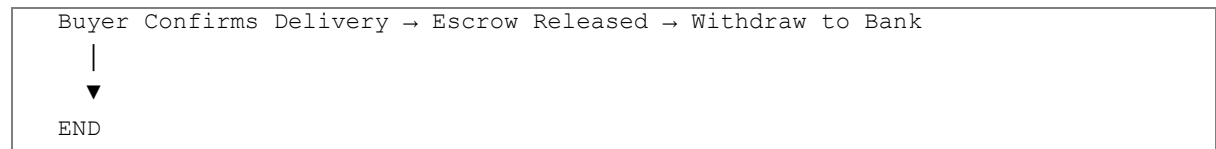


Figure 7.2: Farmer (Seller) — Whole Lot Auction User Flow

7.2.2 Buyer — Partial Order Flow



Figure 7.3: Buyer — Partial Order User Flow

7.2.3 Buyer — RFQ Flow

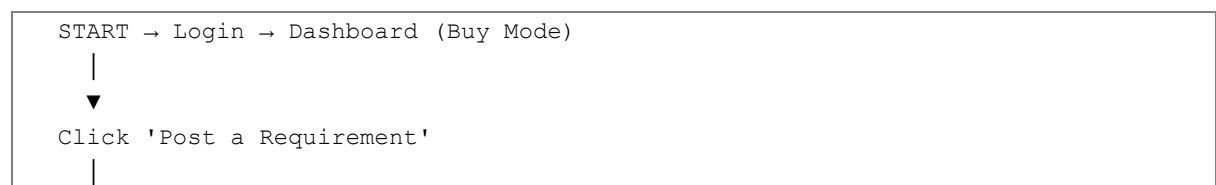




Figure 7.4: Buyer — RFQ (Request for Quotation) User Flow

7.2.4 Dispute Resolution Flow

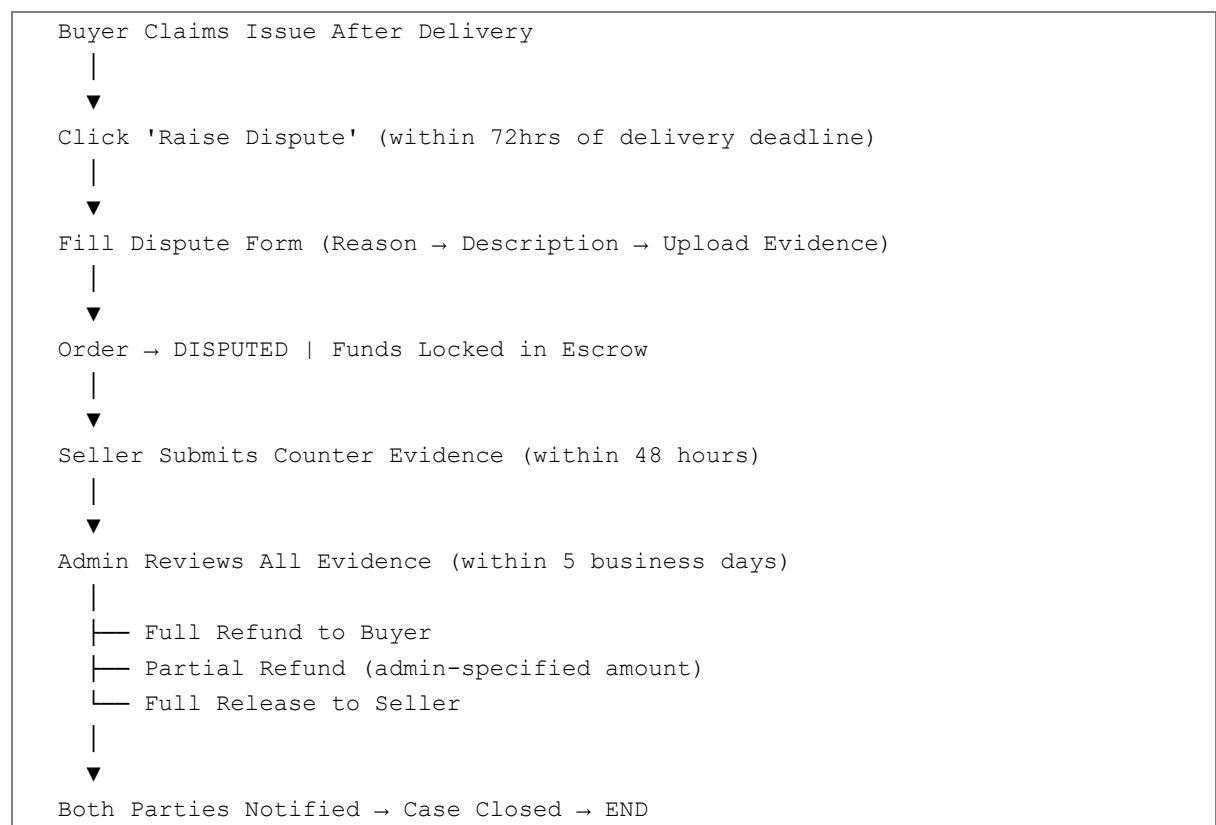


Figure 7.5: Dispute Resolution User Flow

7.3 Design System & Component Library

The KisanSetu Design System ensures visual consistency across all screens and modules. It is built on an 8px base spacing grid and follows WCAG 2.1 AA accessibility standards.

7.3.1 Color Palette

Role	Color Name	Hex Code	Usage
Primary	KisanGreen	#2E7D32	Primary buttons, active states, links
Primary Light	Light Green	#A5D6A7	Hover states, card backgrounds
Secondary	EarthBrown	#795548	Secondary actions, borders
Accent	HarvestGold	#F9A825	Highlights, auction bid updates
Success	Success Green	#388E3C	Confirmation messages, escrow released
Warning	Warning Orange	#F57C00	Countdown timers (mid), pending states
Danger	Danger Red	#D32F2F	Error messages, disputes, low timer
Neutral Dark	Charcoal	#212121	Primary body text
Neutral Light	Off White	#F5F5F5	Page and card backgrounds
White	Pure White	#FFFFFF	Card surfaces, modal backgrounds

Table 7.1: KisanSetu Color Palette and Design System Colors

7.3.2 Typography

Style	Font	Size	Weight	Usage
Heading 1	Inter	32px	Bold (700)	Page titles
Heading 2	Inter	24px	SemiBold (600)	Section headers
Heading 3	Inter	20px	SemiBold (600)	Card titles
Body Large	Inter	16px	Regular (400)	Primary body text

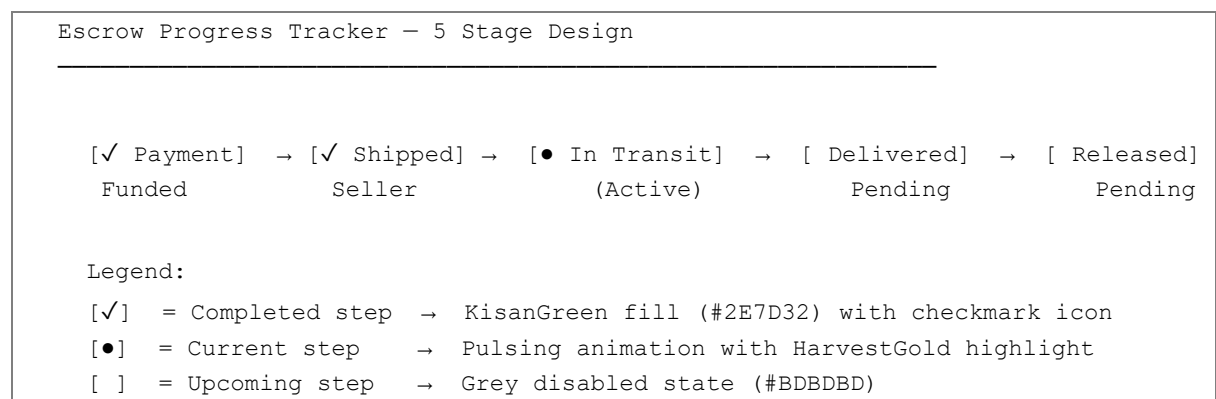
Body Small	Inter	14px	Regular (400)	Secondary text
Caption	Inter	12px	Regular (400)	Labels, timestamps, badges
Button Text	Inter	14px	SemiBold (600)	All button labels

Table 7.2: Typography Scale — Font Sizes, Weights, and Usage**7.3.3 Status Badges**

Badge Label	Color	Used For
ACTIVE	Green (#2E7D32)	Live listings and auctions
PENDING REVIEW	Gold (#F9A825)	Awaiting admin approval
AUCTION ENDED	Grey (#757575)	Completed auction period
OUT OF STOCK	Red (#D32F2F)	Fully ordered listings
CONFIRMED	Green (#388E3C)	Confirmed orders
DISPUTED	Orange (#F57C00)	Orders under dispute
COMPLETED	Dark Green (#1B5E20)	Successfully closed orders
CANCELLED	Red (#D32F2F)	Cancelled orders or listings

Table 7.3: Status Badge Labels, Colors, and Usage**7.3.4 Escrow Progress Tracker**

A horizontal step indicator with 5 stages used on all order detail pages (Figure 7.6):

**Figure 7.6: Escrow Progress Tracker — Step Indicator Design**

7.4 Wireframe to High-Fidelity Design Process

Stage 1 — Research & Problem Definition:

User personas were defined based on the PRD — Ramnath (small farmer), Sunita (progressive farmer), Vikram (wholesale buyer), Arjun (procurement manager). Pain points from each persona informed the priority of UI screens.

Stage 2 — Information Architecture:

A complete sitemap was created mapping all screens, their relationships, and access levels (public / authenticated / admin) before any visual design began.

Stage 3 — Low-Fidelity Wireframes:

Rough wireframes were sketched for all critical flows — registration, dashboard, listing creation, auction bidding, partial order checkout, escrow tracker, RFQ, and admin panel. Layout and content hierarchy were established without any color or styling.

Stage 4 — Design System Creation:

The color palette, typography scale, spacing grid, and core component library were defined in Figma before high-fidelity screens were built, ensuring consistency across all 30+ screens.

Stage 5 — High-Fidelity UI Design in Figma:

All screens were designed at full fidelity in Figma using the established design system. Auto-layout was used throughout for responsive behavior. Components were created as Figma instances linked to the master component library.

Stage 6 — Interactive Prototype:

Key user flows were connected in Figma's prototype mode — including the full auction bidding flow, partial order checkout with countdown timer, and the escrow status progression — and published to the Figma site at kisansetu.figma.site.

CHAPTER 8: PROJECT SCREENSHOTS

8.1 Login UI

The figure displays two screenshots of the KisanSetu web application interface, overlaid on a background image of a farmer in a yellow turban holding a tablet in a field with a tractor.

Top Screenshot: Join KisanSetu (Registration)

- Header:** Home, Features, Market Rates, Schemes, Login, Sign Up.
- Modal Title:** Join KisanSetu. Subtext: Create your account and start your journey.
- Form Fields:**
 - Full Name: Enter full name
 - Phone Number: +91 98765 43210
 - Email Address: your.mail@example.com
 - Password: Create password
 - Confirm Password: Re-enter password
- Agreement:** ☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)
- Buttons:** Create Account (green), OR, Continue with Google (Google logo).
- Footer:** Already have an account? [Login](#)

Bottom Screenshot: Welcome Back (Login)

- Header:** Home, Features, Market Rates, Schemes, Login, Sign Up.
- Modal Title:** Welcome Back. Subtext: Login to access your KisanSetu account.
- Form Fields:**
 - Email Address: your.email@example.com
 - Password: Enter your password
- Options:** ☐ Remember me, [Forgot Password?](#)
- Buttons:** Login (green), OR, Continue with Google (Google logo).
- Footer:** Don't have an account? [Sign Up](#)

Figure 8.1: KisanSetu Login & Registration Screen

8.2 Dashboard UI (Sell Mode)

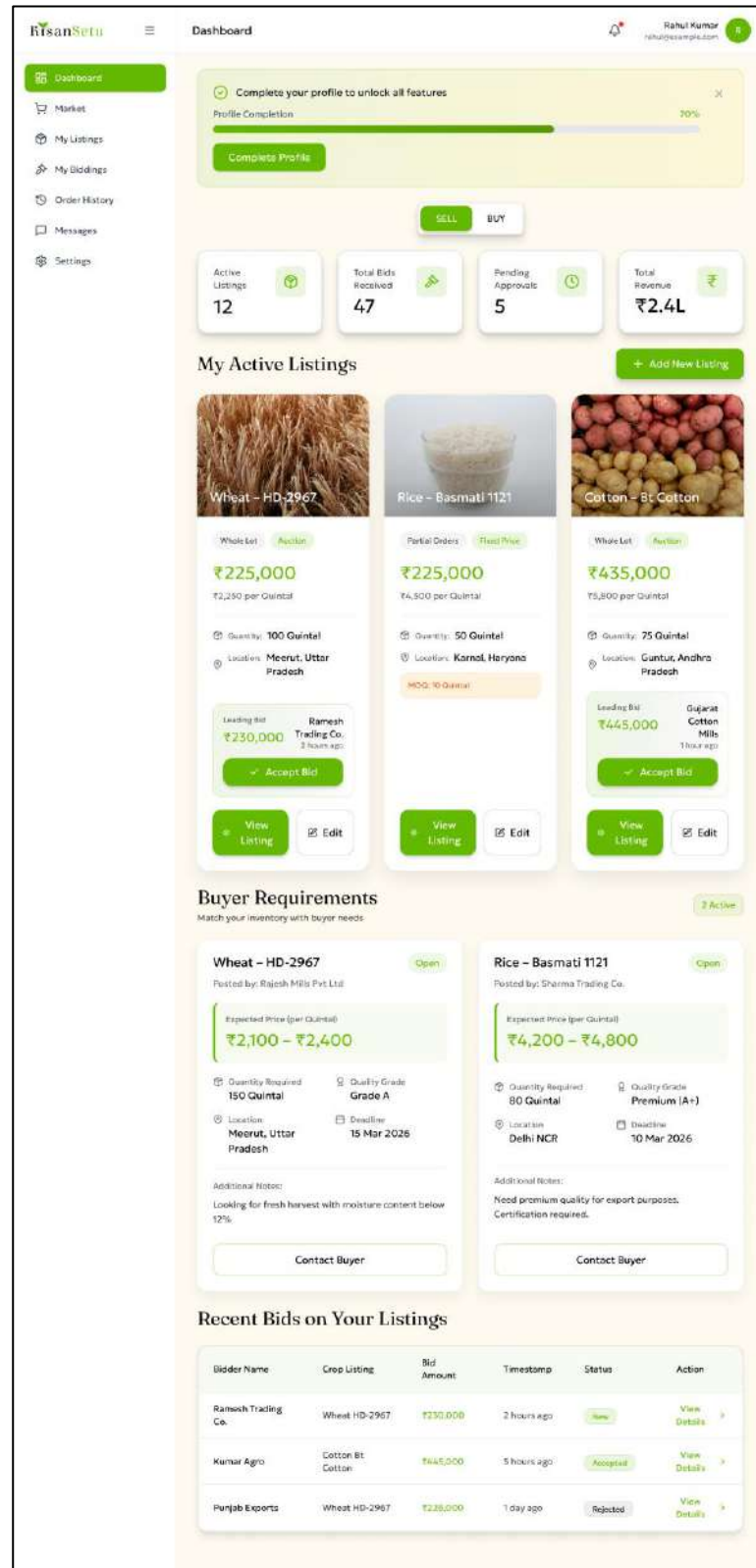


Figure 8.2: Dashboard UI — Sell Mode (Farmer View)

8.3 Dashboard UI (Buy Mode)

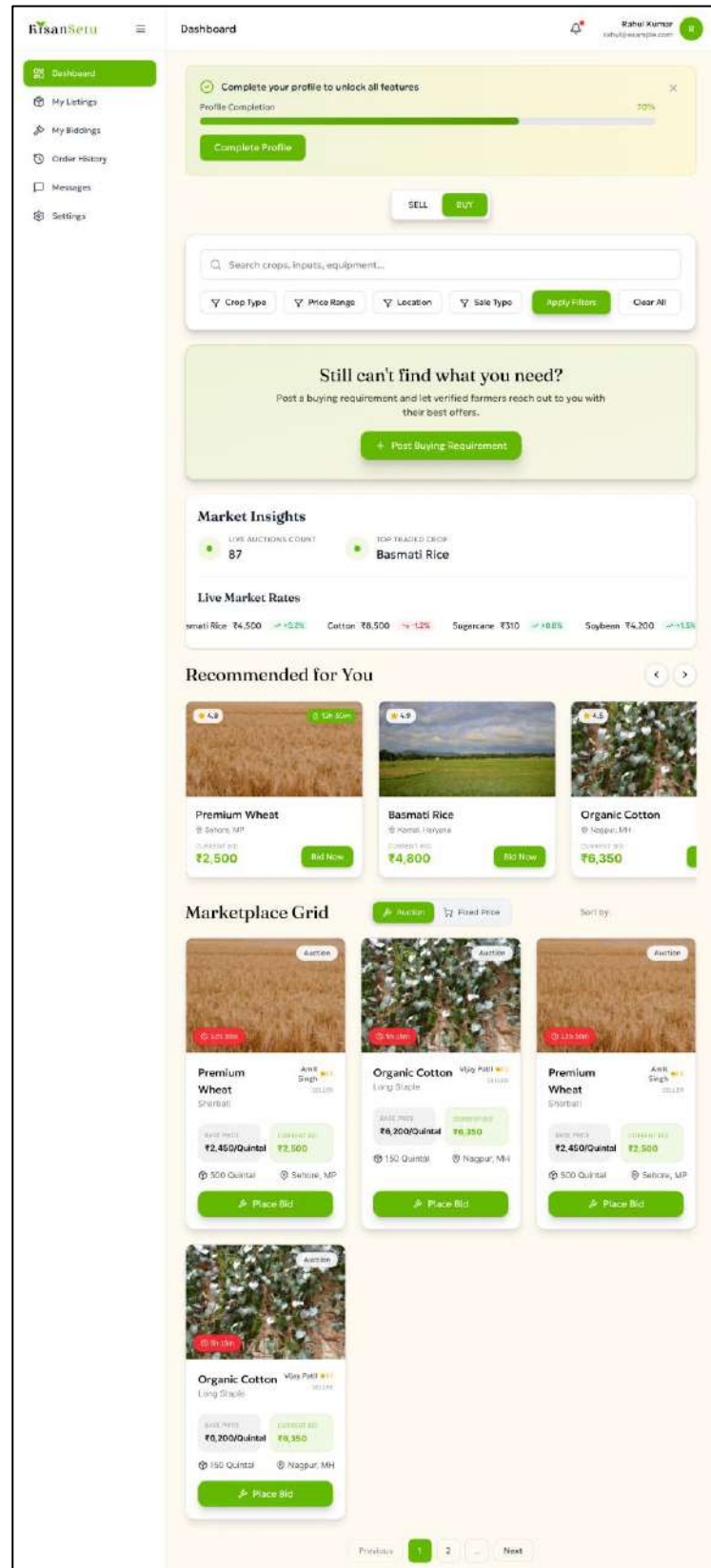


Figure 8.3: Dashboard UI — Buy Mode (Buyer View)

8.4 Create Listing UI

The screenshot shows the 'Create New Listing' form in the 'Product Details' step. The form is part of a 5-step process: 1. Product Details, 2. Quantity & Pricing, 3. Purchase Type, 4. Quality & Location, and 5. Media & Review. The 'Product Details' step is currently active. The form includes fields for 'Crop Name' (a dropdown menu), 'Variety' (a text input field), 'Grade' (a dropdown menu), and 'Harvest Date' (a date picker). At the bottom, there are buttons for 'Back', 'Save as Draft', and 'Next'. Below the form, a table displays three listing examples with their respective prices and quantities.

Listing	Price	Quantity
1	₹225,000	₹2,250 per Quintal
2	₹225,000	₹4,500 per Quintal
3	₹435,000	₹5,800 per Quintal

The screenshot shows the 'Create New Listing' form in the 'Quantity & Sale Details' step. The form is part of a 5-step process: 1. Product Details, 2. Quantity & Pricing, 3. Purchase Type, 4. Quality & Location, and 5. Media & Review. The 'Quantity & Pricing' step is currently active. The form includes fields for 'Total Quantity' (a text input field), 'Unit' (a dropdown menu), 'Sale Type' (radio buttons for 'Auction' and 'Fixed Price'), 'Price Per kg' (a text input field), 'Minimum Bid Increment' (a text input field), and 'Auction End Date' (a date picker). At the bottom, there are buttons for 'Back', 'Save as Draft', and 'Next'.

Create New Listing

List your crop for auction or fixed price sale

1

2

3

4

5

Product Details

Quantity & Pricing

Purchase Type

Quality & Location

Media & Review

Purchase Type

How would you like buyers to purchase your crop? *

☒ Whole Lot Only

Buyers must purchase the entire quantity you're listing. This is ideal when you want to sell all your produce in a single transaction. Example: If you list 1000 kg, the buyer must buy all 1000 kg.

☐ Partial Orders Allowed

Buyers can purchase any quantity they need, as long as it meets your minimum order quantity (MOQ). This allows multiple buyers to purchase from your listing. Example: If you list 1000 kg with MOQ of 100 kg, buyers can order 100 kg, 250 kg, 500 kg, etc.

< Back

Save as Draft

Next

KisanSetu

Dashboard

Rahul Kumar
rahul@example.com

Create New Listing

List your crop for auction or fixed price sale

1

2

3

4

5

Product Details

Quantity & Pricing

Purchase Type

Quality & Location

Media & Review

Quality & Logistics

Packaging Type *

Select Packaging

Storage Type

Select Storage Type

State *

Select State

District *

Select District

Pickup Method *

☒ Buyer Pickup

☐ Seller Delivery

☐ Transport Negotiable

< Back

Save as Draft

Next

KisanSetu

Dashboard

Rahul Kumar
rahul@example.com

Create New Listing

List your crop for auction or fixed price sale

1

Product Details

2

Quantity & Pricing

3

Purchase Type

4

Quality & Location

5

Media & Review

Media & Additional Details

Product Images

Drag & Drop Images Here
or click to browse

Choose Files

Upload Quality Certificate (Optional)

PDF or Image files accepted

Choose File

Description (Optional)

Mention storage condition, certification, or additional notes...

Listing Preview

Crop Name - Variety

Whole Lot

Quantity: --- kg

Sale Type: Auction

Price (Total): ₹---

Per kg: ₹---

Location: ---, ---

< Back

Save as Draft

Publish Listing

Figure 8.4: Create Listing — Multi-Step Form UI

8.5 Auction Bidding UI

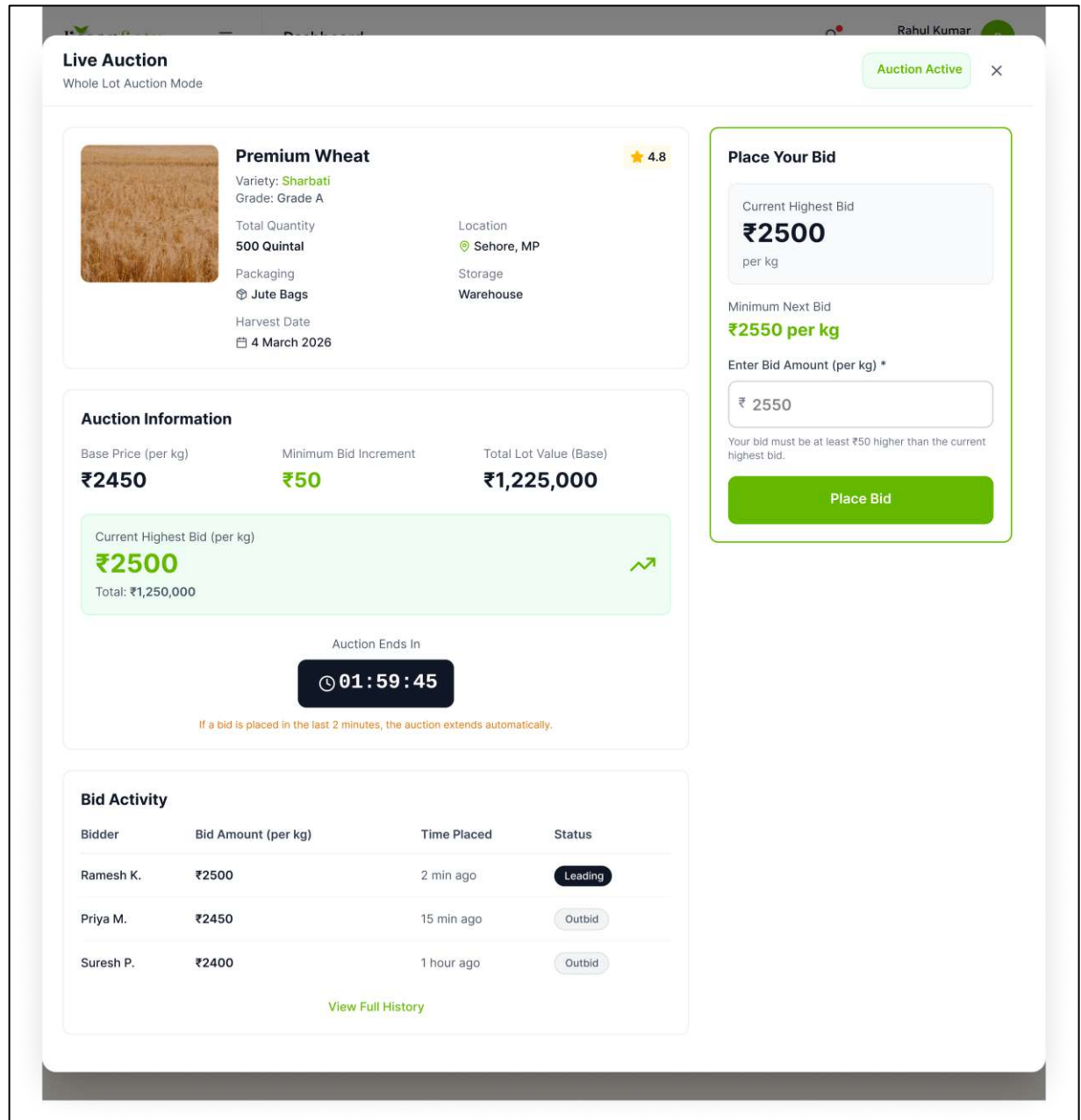


Figure 8.5: Auction Bidding Interface with Live Bid Counter

8.6 Partial Order UI

Fixed Price Purchase

Whole Lot or Partial Orders Available

Available

X



Basmati Rice

Variety: **Pusa 1121**

Grade: **Grade A**

Total Quantity
200 Quintal

Packaging
Jute Bags

Harvest Date
4 March 2025

Location
Karnal, Haryana

Storage
Warehouse

★ 4.9

Pricing Information

Fixed Price per Quintal
₹4800

Total Price (Whole Lot)
₹960,000

Partial Orders Available

Minimum Order Quantity: **20 Quintal**

Price for partial orders: **₹4750 per Quintal**

Purchase Now

Fixed Price
₹4800
per Quintal

Total (Whole Lot)
₹960,000

Available Quantity
200 Quintal

Flexible Ordering

You can choose to buy the whole lot or order a partial quantity (min. 20 Quintal)

Buy Now

Contact Seller

Confirm Purchase

Please review your order details before confirming

Order Type
Partial Order

Product
Basmati Rice - Pusa 1121

Quantity (Quintal)

Min: 20 Quintal • Max: 200 Quintal

Price per Quintal: ₹4750

Quantity: 20 Quintal

Total Amount: **₹95,000**

☒ Enable Escrow Protection

Funds will be held securely until delivery is confirmed

Cancel

Confirm Purchase

Figure 8.6: Partial Order Checkout

8.7 Escrow Status Tracker UI

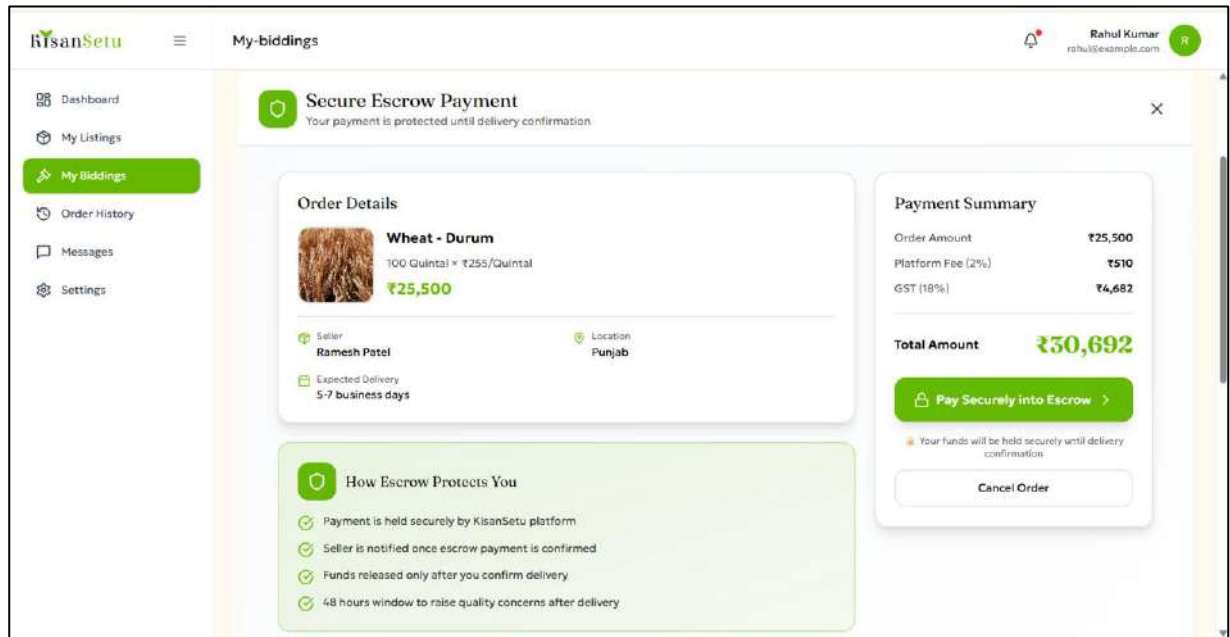
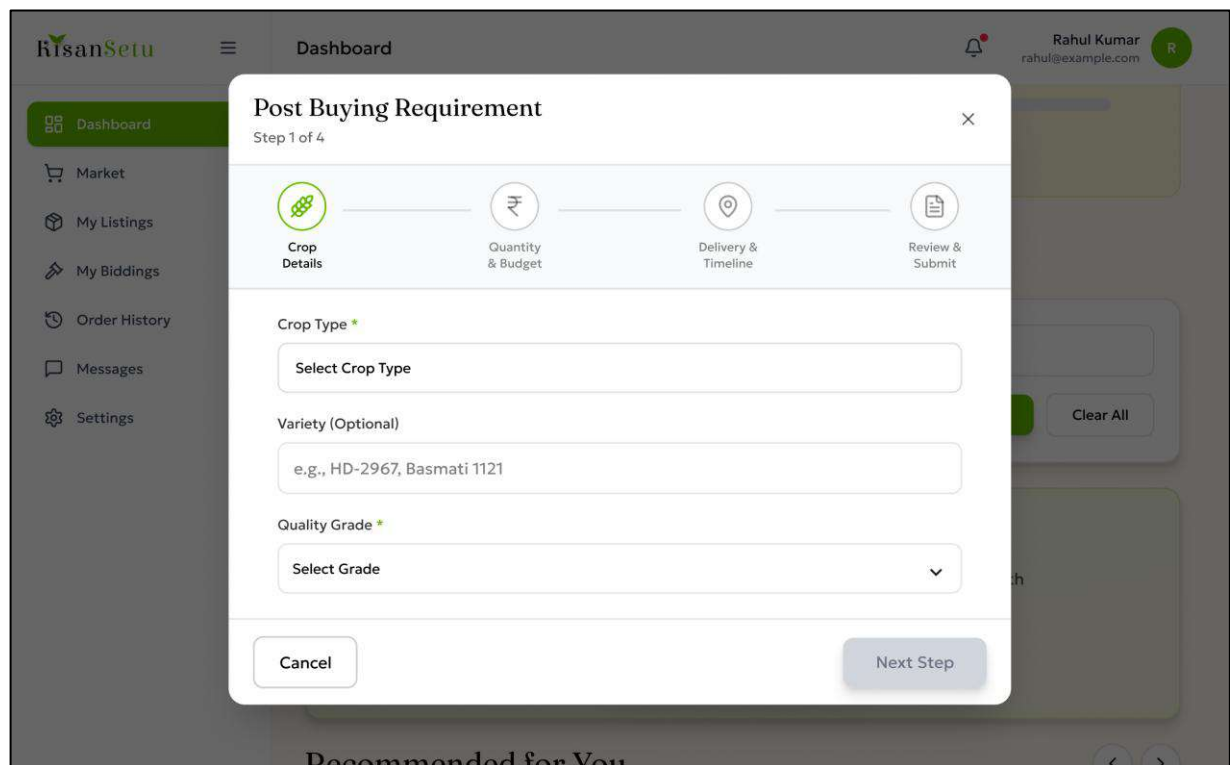


Figure 8.7: Escrow Status Tracker Screen

8.8 RFQ / Buyer Requirement UI



KisanSetu Dashboard

Rahul Kumar
rahul@example.com

Post Buying Requirement
Step 2 of 4

Progress: Crop Details → **Quantity & Budget** → Delivery & Timeline → Review & Submit

Quantity Required *
Enter quantity Quintal

Expected Price Range (per quintal) *
Min price (₹) Max price (₹)

Back Next Step

KisanSetu Dashboard

Rahul Kumar
rahul@example.com

Post Buying Requirement
Step 3 of 4

Progress: Crop Details → Quantity & Budget → **Delivery & Timeline** → Review & Submit

State *
e.g., Uttar Pradesh

District *
e.g., Meerut

Delivery Location (Full Address) *
Complete delivery address

Deadline *
dd-mm-yyyy

Urgency
Normal Urgent

Back Next Step

KisanSetu

Post Buying Requirement

Step 4 of 4

Crop Details | Quantity & Budget | Delivery & Timeline | Review & Submit

Additional Notes

Add any specific requirements, quality standards, or special instructions...

Requirement Summary

Crop:	Wheat - gt
Quality:	Premium (A+)
Quantity:	20 kg
Price Range:	₹30 - ₹60 per kg
Location:	meerut, up
Deadline:	26 March 2026
Urgency:	Normal

< Back | Submit Requirement

Figure 8.8: RFQ Posting and Offer Comparison Table

8.9 Messaging / Chat UI

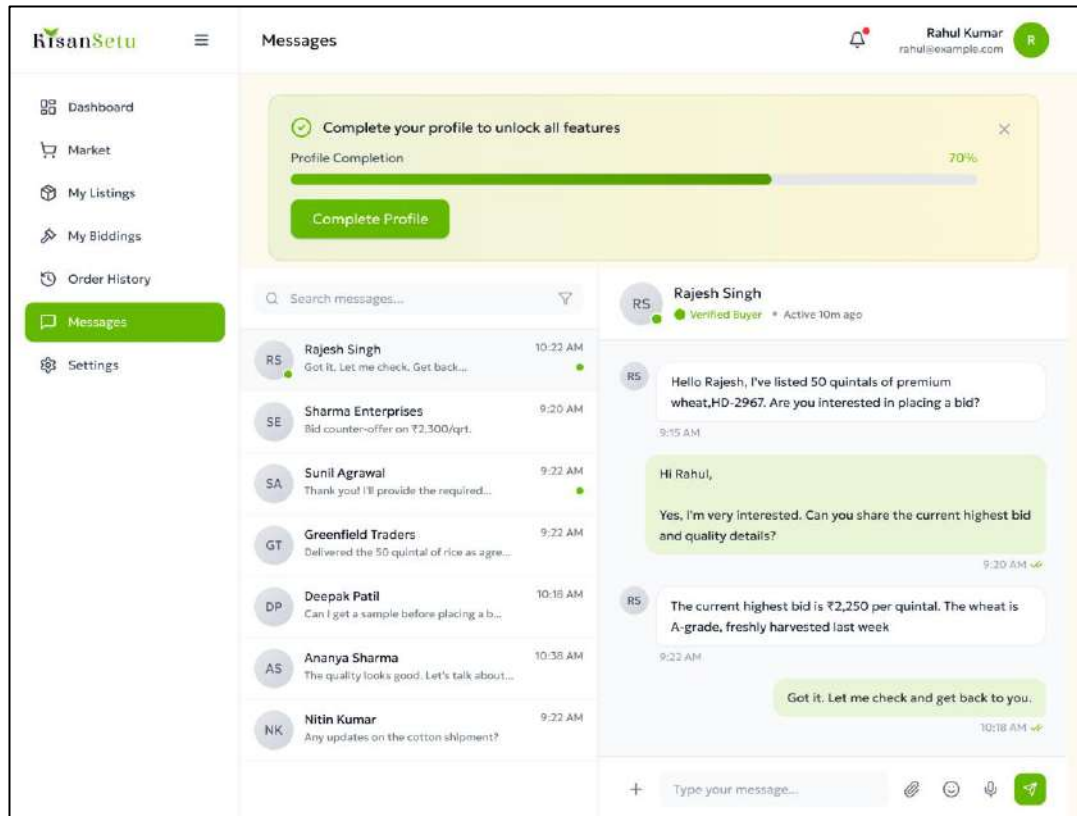
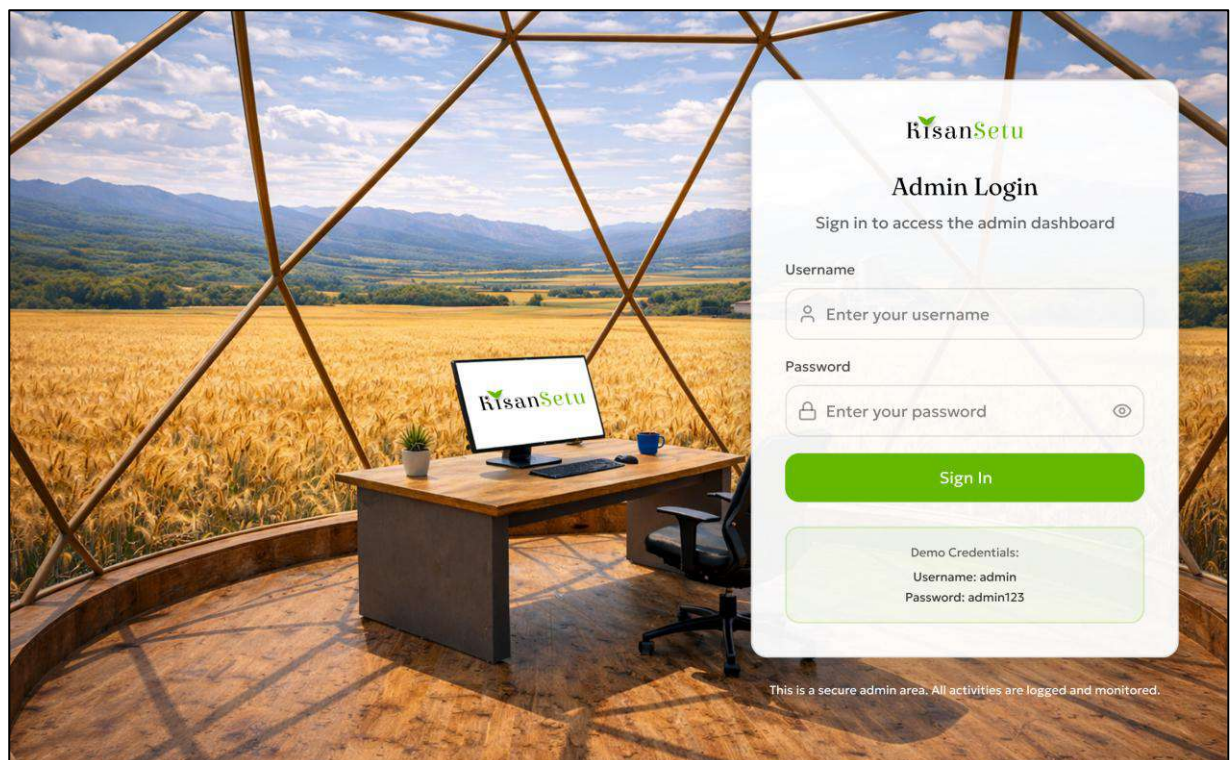


Figure 8.9: In-Platform Messaging / Order Chat Screen

8.10 Admin Panel UI



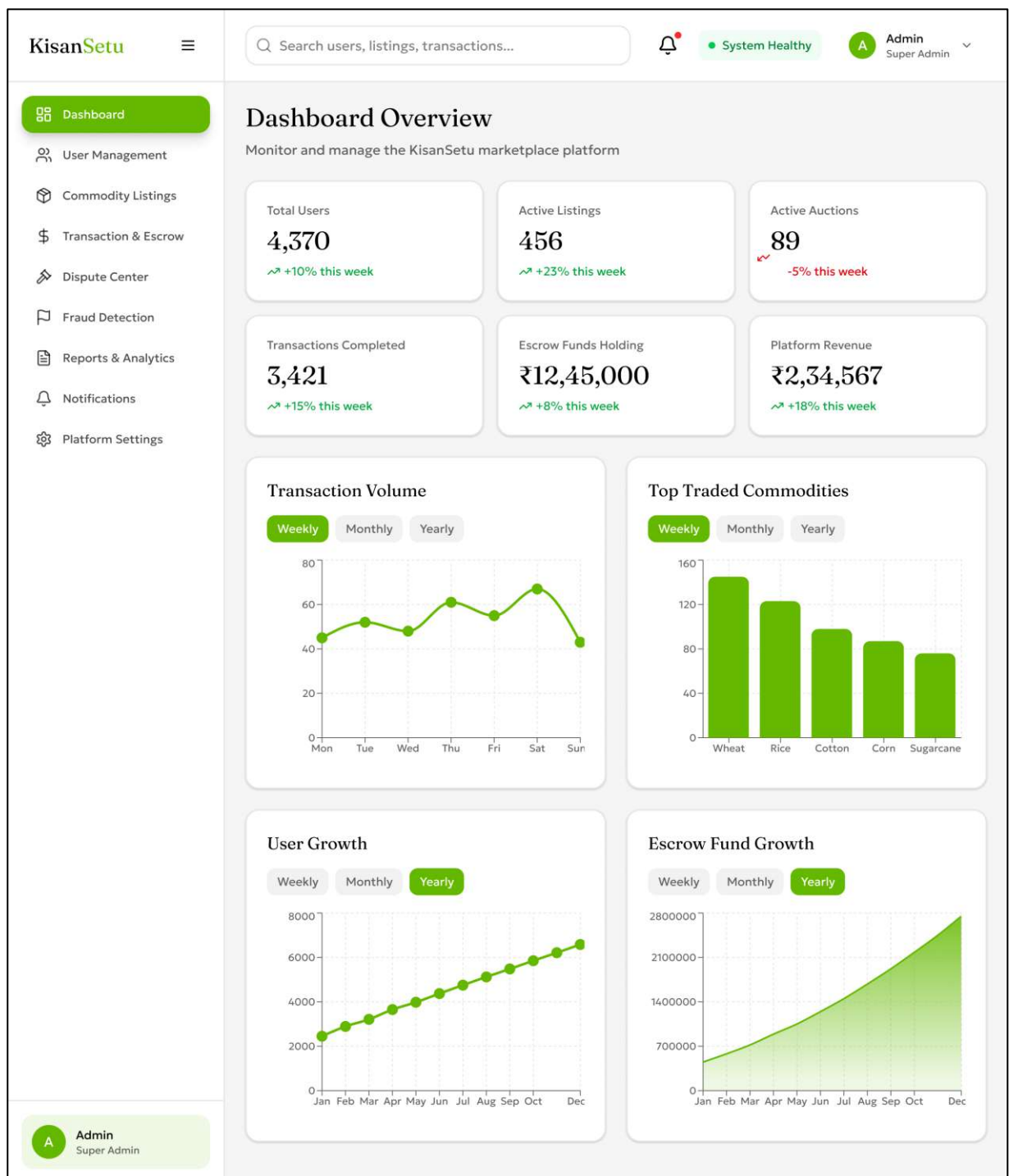


Figure 8.10: Admin Panel — Listing Moderation Queue

CHAPTER 9: CONCLUSION AND DISCUSSION

9.1 Overall Analysis of Internship

My internship at **24×7 Moneyworks Consulting Pvt. Ltd.** has been a highly enriching and transformative experience, providing me with valuable insights and practical exposure to the field of web development. Throughout the internship, I had the opportunity to work on real-world applications and understand the complete development process, from requirement analysis to implementation.

This experience allowed me to work on a live project, the **Farmer Crop Sell Bid System**, where I applied frontend development concepts such as component-based architecture, API integration, state management, and responsive design using technologies like React.js, Next.js, and Redux Toolkit. It helped me understand the importance of usability, performance, and scalability in building modern web applications.

The internship significantly enhanced my technical knowledge, problem-solving skills, and logical thinking abilities. It also improved my communication and collaboration skills through continuous interaction with my internal guide **Mr. Paresh Patel**, mentors, and team members. I gained a clear understanding of how frontend development integrates with backend systems to deliver efficient and user-friendly solutions.

Overall, this internship has played a crucial role in shaping my career direction in the field of Information Technology and has prepared me for future opportunities in software and web development.

9.2 Dates of continuous evaluation (CE-I and CE-II)

Evaluation	Date	Remarks
CE – I	5th February 2026	Conducted by internal guide Prof. Paresh Patel. Focus on initial development work, understanding of frontend concepts, and progress in project implementation along with planning for upcoming modules.
CE – II	4th April 2026	Conducted by internal guide Mr. Paresh Patel. Focus on evaluation of project modules, API integration, UI development, and overall learning outcomes.

Table 9.1: Continuous Evaluation Dates

9.3 Problem encountered and possible solutions

Problem Encountered	Solution Applied
Handling real-time bid updates in the application	Implemented efficient API handling and optimized UI updates for displaying latest bid values
Managing complex state across multiple components	Used Redux Toolkit for centralized state management and better data flow
API integration and error handling issues	Implemented proper error handling and validation for smooth user experience
Designing responsive UI for different devices	Applied responsive design techniques to ensure compatibility across mobile, tablet, and desktop
Form validation and data handling in listing module	Used structured validation and controlled inputs to prevent incorrect data submission
Managing multiple user roles (buyer and seller)	Implemented role-based functionality within a single system for better usability
Performance optimization for large data handling	Optimized rendering and reduced unnecessary API calls

Table 9.2: Problems Encountered and 9.2 Solutions Applied

9.4 Summary of internship

Throughout my internship at 24×7 Moneyworks Consulting Pvt. Ltd., I gained valuable experience in the field of web development. This internship provided both theoretical knowledge and practical exposure, helping me understand how real-world applications are developed and maintained.

The internship work focused on developing the Farmer Crop Sell Bid System, which involved building user interfaces, integrating APIs, implementing authentication systems, and managing application state. This hands-on experience helped me apply development concepts in real-world scenarios and improve my coding skills.

One of the key highlights of the internship was working with modern technologies such as React.js, Next.js, and Redux Toolkit. These tools helped me understand application structure, performance optimization, and scalable development practices.

Overall, the internship was highly beneficial, providing a strong foundation in frontend development and enhancing my technical, analytical, and problem-solving skills. It has prepared me for future opportunities in the IT industry.

9.5 Limitation and future enhancement

There are still some improvements and optimizations required to enhance the overall performance and functionality of the project. Currently, the developed system represents a portion of a larger real-time application.

In the future, additional features and modules can be implemented based on requirements, such as real-time bidding using WebSocket, improved security measures, enhanced user interface, and better performance optimization. Further development will focus on scalability, advanced features, and improved user experience.

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